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## **1. Introduction**

In Progress Report 3, I configured the WAN using Cisco Packet Tracer and connected the company networks such as Company A, Company B, and Company C. In addition to that, I have also finished configuring VLAN for Company A, since in the last progress report, I have finished the DMZ network and the internal server network for Company A but left the rest for Company A to complete later using VLAN.

The main objectives that were achieved in this progress report include the following:

Completion of LAN designing using VLAN technology for Company A

Configuration of the WAN to interconnect Company A, Company B, and Company C

Routing configuration between the different networks of different companies.

## **2. VLAN and WAN**

In Progress Report 2, the basic network topology design of Company A was already developed on Packet Tracer through the use of routers, switches, DMZ switch, internal switch, and access point. During that time, the DMZ subnet and internal server subnet have already been tested and succeeded. However, the other departmental networks were supposed to be done afterwards through VLANs.

In this progress report, the other departmental networks of Company A were successfully implemented through VLANs and router subinterfaces. The WAN was also successfully completed through the interconnection of Companies A, B, and C via static routing.

## **3. Company A VLAN**

Company A uses these subnets:

Engineering: 216.244.168.0/25

Manufacturing: 216.244.168.128/25

Sales: 216.244.169.0/26

Administration: 216.244.169.64/27

Internal Server: 216.244.169.96/28

DMZ: 216.244.169.112/29

Guest WLAN: 216.244.169.128/26

WAN: 216.244.175.252/30

```
Switch#show vlan brief
```

| VLAN Name               | Status | Ports                                                                                                                                                       |
|-------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 default               | active | Fa0/7, Fa0/8, Fa0/9, Fa0/10<br>Fa0/11, Fa0/12, Fa0/13, Fa0/14<br>Fa0/15, Fa0/16, Fa0/17, Fa0/18<br>Fa0/19, Fa0/20, Fa0/21, Fa0/22<br>Fa0/23, Gig0/1, Gig0/2 |
| 10 ENGINEERING          | active | Fa0/2                                                                                                                                                       |
| 20 MANUFACTURING        | active | Fa0/3                                                                                                                                                       |
| 30 SALES                | active | Fa0/4                                                                                                                                                       |
| 40 ADMIN                | active | Fa0/5                                                                                                                                                       |
| 50 INTERNAL_SERVER      | active | Fa0/1                                                                                                                                                       |
| 70 GUEST_WLAN           | active | Fa0/6                                                                                                                                                       |
| 1002 fddi-default       | active |                                                                                                                                                             |
| 1003 token-ring-default | active |                                                                                                                                                             |
| 1004 fddinet-default    | active |                                                                                                                                                             |
| 1005 trnet-default      | active |                                                                                                                                                             |

```
Switch#
```

```
Router>show ip interface brief
```

| Interface             | IP-Address      | OK? | Method | Status                | Protocol |
|-----------------------|-----------------|-----|--------|-----------------------|----------|
| GigabitEthernet0/0    | 216.244.169.113 | YES | manual | up                    | up       |
| GigabitEthernet0/1    | unassigned      | YES | manual | up                    | up       |
| GigabitEthernet0/1.10 | 216.244.168.1   | YES | manual | up                    | up       |
| GigabitEthernet0/1.20 | 216.244.168.129 | YES | manual | up                    | up       |
| GigabitEthernet0/1.30 | 216.244.169.1   | YES | manual | up                    | up       |
| GigabitEthernet0/1.40 | 216.244.169.65  | YES | manual | up                    | up       |
| GigabitEthernet0/1.50 | 216.244.169.97  | YES | manual | up                    | up       |
| GigabitEthernet0/1.70 | 216.244.169.129 | YES | manual | up                    | up       |
| GigabitEthernet0/2    | 216.244.175.254 | YES | manual | up                    | up       |
| Vlan1                 | unassigned      | YES | unset  | administratively down | down     |

## 4. Routing Method

For the WAN routing, we considered static routing, RIPv2, and OSPFv2.

### Static Routing:

#### Advantages

- simple
- easy to control
- good for a small network

#### Disadvantages

- routes must be added manually
- not good for a large network

### RIPv2:

#### Advantages

- automatically shares routes
- easier than OSPF

#### Disadvantages

Slower

not as efficient for a bigger WAN

### **OSPFv2:**

#### **Advantages**

faster

better for larger networks

#### **Disadvantages**

more complicated to configure

Selected Method

For this project, we used static routing. we chose it because the WAN is small and it is easier to manage with manual routes.

## **5. WAN Topology**

The WAN was built using one central router called R-WAN. This router connects the company routers of Company A, Company B, and Company C.

The WAN connections are:

- Company A router to R-WAN
- Company B router to R-WAN
- Company C router to R-WAN

#### **Company A WAN**

Company A router interface: G0/2

R-WAN interface: G0/0

Subnet: 216.244.175.252/30

Company A router IP: 216.244.175.254

R-WAN IP: 216.244.175.253

#### **Company B WAN**

Company B router interface: G0/2

R-WAN interface: G0/2

Subnet: 216.244.183.252/30

Company B router IP: 216.244.183.254

R-WAN IP: 216.244.183.253

#### **Company C WAN**

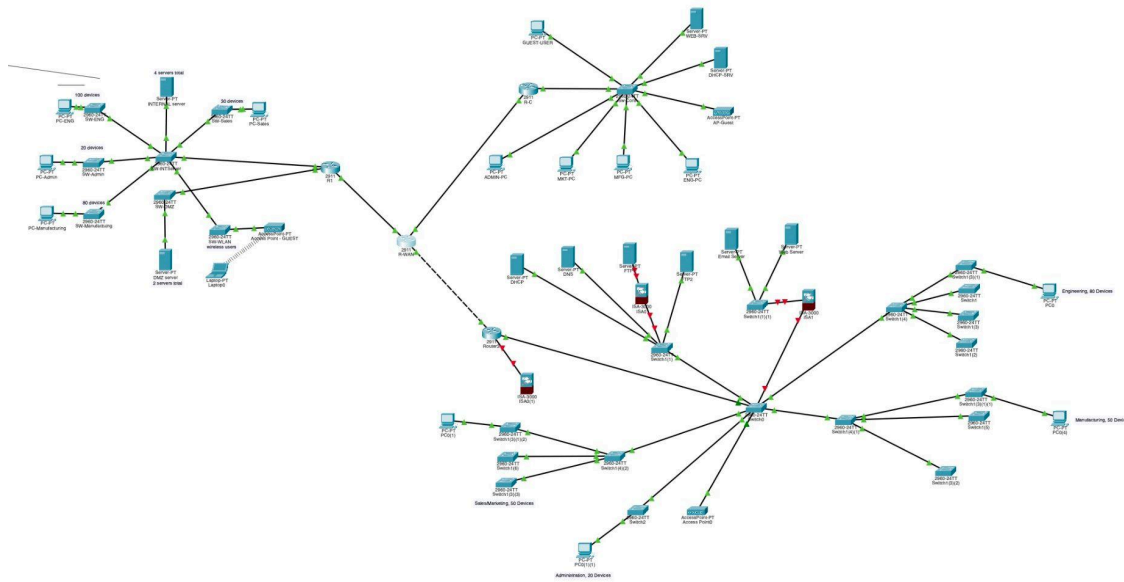
Company C router interface: G0/1

R-WAN interface: G0/1

Subnet: 216.244.254.252/30

Company C router IP: 216.244.254.254

R-WAN IP: 216.244.254.253



```

Router>show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  216.244.175.253 YES manual  up          up
GigabitEthernet0/1  216.244.254.253 YES manual  up          up
GigabitEthernet0/2  216.244.183.253 YES manual  up          up
Vlan1              unassigned      YES unset   administratively down down

```

## 6. WAN Configuration

After the WAN links were connected, static routes were configured so that the three companies could communicate.

R-WAN has routes for Company A, Company B, and Company C.

Company A router has routes for Company B and Company C.

Company B router uses a default route to R-WAN.

Company C router has routes for Company A and Company B.

### R-WAN

```

Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    216.244.168.0/21 [1/0] via 216.244.175.254
    216.244.175.0/24 is variably subnetted, 2 subnets, 2 masks
C    216.244.175.252/30 is directly connected, GigabitEthernet0/0
L    216.244.175.253/32 is directly connected, GigabitEthernet0/0
S    216.244.176.0/23 [1/0] via 216.244.183.254
    216.244.183.0/24 is variably subnetted, 2 subnets, 2 masks
C    216.244.183.252/30 is directly connected, GigabitEthernet0/2
L    216.244.183.253/32 is directly connected, GigabitEthernet0/2
S    216.244.184.0/23 [1/0] via 216.244.254.254
    216.244.254.0/24 is variably subnetted, 2 subnets, 2 masks
C    216.244.254.252/30 is directly connected, GigabitEthernet0/1
L    216.244.254.253/32 is directly connected, GigabitEthernet0/1

```

## Company A

```

Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

```

Gateway of last resort is not set

```

    216.244.168.0/24 is variably subnetted, 4 subnets, 2 masks
C    216.244.168.0/25 is directly connected, GigabitEthernet0/1.10
L    216.244.168.1/32 is directly connected, GigabitEthernet0/1.10
C    216.244.168.128/25 is directly connected, GigabitEthernet0/1.20
L    216.244.168.129/32 is directly connected, GigabitEthernet0/1.20
    216.244.169.0/24 is variably subnetted, 10 subnets, 5 masks
C    216.244.169.0/26 is directly connected, GigabitEthernet0/1.30
L    216.244.169.1/32 is directly connected, GigabitEthernet0/1.30
C    216.244.169.64/27 is directly connected, GigabitEthernet0/1.40
L    216.244.169.65/32 is directly connected, GigabitEthernet0/1.40
C    216.244.169.96/28 is directly connected, GigabitEthernet0/1.50
L    216.244.169.97/32 is directly connected, GigabitEthernet0/1.50
C    216.244.169.112/29 is directly connected, GigabitEthernet0/0
L    216.244.169.113/32 is directly connected, GigabitEthernet0/0
C    216.244.169.128/26 is directly connected, GigabitEthernet0/1.70
L    216.244.169.129/32 is directly connected, GigabitEthernet0/1.70
    216.244.175.0/24 is variably subnetted, 2 subnets, 2 masks
C    216.244.175.252/30 is directly connected, GigabitEthernet0/2
L    216.244.175.254/32 is directly connected, GigabitEthernet0/2
S    216.244.176.0/23 [1/0] via 216.244.175.253
    216.244.183.0/30 is subnetted, 1 subnets
S    216.244.183.252/30 [1/0] via 216.244.175.253
S    216.244.184.0/23 [1/0] via 216.244.175.253

```

## Company B

```

Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 216.244.183.253 to network 0.0.0.0

    216.244.176.0/24 is variably subnetted, 6 subnets, 3 masks
C       216.244.176.0/25 is directly connected, GigabitEthernet0/0.10
L       216.244.176.1/32 is directly connected, GigabitEthernet0/0.10
C       216.244.176.128/26 is directly connected, GigabitEthernet0/0.20
L       216.244.176.129/32 is directly connected, GigabitEthernet0/0.20
C       216.244.176.192/26 is directly connected, GigabitEthernet0/0.30
L       216.244.176.193/32 is directly connected, GigabitEthernet0/0.30
    216.244.177.0/24 is variably subnetted, 6 subnets, 4 masks
C       216.244.177.0/27 is directly connected, GigabitEthernet0/0.40
L       216.244.177.1/32 is directly connected, GigabitEthernet0/0.40
C       216.244.177.32/29 is directly connected, GigabitEthernet0/0.50
L       216.244.177.33/32 is directly connected, GigabitEthernet0/0.50
C       216.244.177.40/30 is directly connected, GigabitEthernet0/0.60
L       216.244.177.41/32 is directly connected, GigabitEthernet0/0.60
    216.244.183.0/24 is variably subnetted, 2 subnets, 2 masks
C       216.244.183.252/30 is directly connected, GigabitEthernet0/2
L       216.244.183.254/32 is directly connected, GigabitEthernet0/2
S*    0.0.0.0/0 [1/0] via 216.244.183.253

```

**Company C**

```

Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    216.244.168.0/21 [1/0] via 216.244.254.253
S    216.244.176.0/23 [1/0] via 216.244.254.253
     216.244.183.0/30 is subnetted, 1 subnets
S    216.244.183.252/30 [1/0] via 216.244.254.253
     216.244.184.0/24 is variably subnetted, 6 subnets, 3 masks
C    216.244.184.0/25 is directly connected, GigabitEthernet0/0.10
L    216.244.184.1/32 is directly connected, GigabitEthernet0/0.10
C    216.244.184.128/26 is directly connected, GigabitEthernet0/0.20
L    216.244.184.129/32 is directly connected, GigabitEthernet0/0.20
C    216.244.184.192/26 is directly connected, GigabitEthernet0/0.30
L    216.244.184.193/32 is directly connected, GigabitEthernet0/0.30
     216.244.185.0/24 is variably subnetted, 8 subnets, 4 masks
C    216.244.185.0/27 is directly connected, GigabitEthernet0/0.40
L    216.244.185.1/32 is directly connected, GigabitEthernet0/0.40
C    216.244.185.32/27 is directly connected, GigabitEthernet0/0.50
L    216.244.185.33/32 is directly connected, GigabitEthernet0/0.50
C    216.244.185.64/28 is directly connected, GigabitEthernet0/0.60
L    216.244.185.65/32 is directly connected, GigabitEthernet0/0.60
C    216.244.185.128/26 is directly connected, GigabitEthernet0/0.70
L    216.244.185.129/32 is directly connected, GigabitEthernet0/0.70
     216.244.254.0/24 is variably subnetted, 2 subnets, 2 masks
C    216.244.254.252/30 is directly connected, GigabitEthernet0/1
L    216.244.254.254/32 is directly connected, GigabitEthernet0/1

```

## 7. Results

### 7.1 Company A to Company C Result

In order to confirm WAN connectivity for Company A, I performed a ping test from the Company A router to the Company C WAN IP address. Ping was successful, proving that Company A had access to Company C through the WAN network.

```

Router#ping 216.244.254.254

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 216.244.254.254, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

```

### 7.2 Company B WAN Result

To test the direct WAN connection to Company B, I did the ping test from R-WAN to the Company B router and it was successful

```
Router>ping 216.244.183.254

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 216.244.183.254, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms
```

### 7.3 Company B to Company A Result

In order to confirm the communication between Company B and Company A, I performed pings on both the Company B router and the WAN IP and LAN Gateway of Company A. The results were successful.

```
Router>ping 216.244.175.254

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 216.244.175.254, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
```

```
Router>ping 216.244.168.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 216.244.168.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

### 7.4 Company B to Company C Result

In order to confirm that there was communication between Company B and Company C, pings were done from the Company B router to the WAN IP of Company C and the LAN Gateway of Company C.

```
Router>ping 216.244.254.254

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 216.244.254.254, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

```
Router>ping 216.244.184.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 216.244.184.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```



The results showed that:

- R-WAN has routes for all three companies,
- Company B router can reach Company A and Company C,
- Company C router can reach Company A and Company B.

This shows that the WAN is working correctly.

## **8. Conclusion**

In Progress Report 3, the company achieved the completion of setting up the VLAN network of the LAN of Company A and setting up the WAN connection between Companies A, B, and C using static routing as the WAN was small in size and easier to manage. From the results achieved by Cisco, there was success in the communication process involving the routers of all the three firms.